THE POSITIVE-MEASURE KAM THEOREM: BLUEPRINT

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Abstract. A complete proof plan, step by step, for the five frozen statements of the KAM formalization: the positive-measure KAM theorem with rate $\varepsilon^{1/4}$ and its qualitative form, the a-posteriori KAM theorem at fixed frequency, local uniqueness of invariant tori of flows, and the measure of the non-Diophantine set. Every result is labelled and cites its pinned source; the dependency graph of the formalization is extracted from this document.

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1. Conventions, definitions and the frozen statements

This part fixes notation and restates, in mathematical form, the five frozen theorems and the twenty frozen definitions of the formalization (owner approval D11, 27 October 2026). The Lean declarations in `KAM/Frozen/` are authoritative; the statements below are their faithful transcriptions, for use by the proofs in Parts 2–5.

1.1. Conventions.

- $n \geq 2$ is the number of degrees of freedom (ruling Q5). Angles $\theta \in \mathbb{R}^n$, actions $I \in \mathbb{R}^n$, phase points $z = (\theta, I)$. Functions are 1-periodic in each angle, so $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ (ruling Q2).
- On $\mathbb{R}^m$ and $\mathbb{C}^m$ we use the max norm $|v| = \max_i |v_i|$; on $\mathbb{C}^n \times \mathbb{C}^n$ the max of the two components. Linear and multilinear maps carry the induced operator norms; for a matrix this is $|M| = \max_i \sum_j |M_{ij}|$ (HL19 §2.1). Sup norms over a set U are written $\|f\|_U = \sup_U |f|$; on a strip, $\|f\|_\rho = \|f\|_{T_\rho}$.
- The complex strip of width $\rho > 0$ is $T_\rho = \{\theta \in \mathbb{C}^n : |\operatorname{Im} \theta_i| < \rho \ \forall i\}$ (HL19 §2.1); it is invariant under translations by $\mathbb{Z}^n$ (and by $\mathbb{R}^n$).
- Fourier characters are $e^{2\pi i k\cdot\theta}$, $k \in \mathbb{Z}^n$, with $\hat{u}_k = \int_{[0,1]^n} u(\theta) e^{-2\pi i k\cdot\theta}\, d\theta$ and average $\langle u \rangle = \hat{u}_0$ (HL19 §2.1). Weighted coefficient norms $\sum_k |\hat{u}_k| e^{|k|_1 s}$ use a *coefficient radius* $s = 2\pi\rho$ (`docs/fourier-conventions.md`).

- Working norms (decision D12). Sup norms on complex domains are the working norms of Parts 3–5, as in HL19 and GJLV05. Weighted coefficient norms enter only through the cohomological equation and Fourier-coefficient arguments of Part 2.

1.2. **Definitions.**

Definition 1.1 (Diophantine vectors)**.** For $\gamma, \tau \in \mathbb{R}$, a vector $\omega \in \mathbb{R}^n$ is (γ,τ)-*Diophantine*, $\omega \in \mathcal{D}_{\gamma,\tau}$, if $\gamma/|k|_1^\tau \le |k\cdot\omega|$ for every $k \in \mathbb{Z}^n \setminus \{0\}$, where $|k|_1 = \sum_i |k_i|$.
Frozen Lean declaration: `KAM.IsDiophantine`
Source: HL19 (3.1), p. 14; Pos01 (1), p. 5.

Definition 1.2 (Complex neighbourhoods and the fundamental domain)**.** For $B \subset \mathbb{R}^n$ and $r > 0$, $B_r = \{I \in \mathbb{C}^n : \exists J \in B,\ |I - J| < r\}$; the phase domain of HEAD is $D(s_0, r_0) = T_{s_0} \times B_{r_0}$. The fundamental domain of $\mathbb{T}^n \times B$ is $F_B = [0,1)^n \times B$, with product Lebesgue measure.
Frozen Lean declaration: `KAM.complexStrip, KAM.actionNbhd, KAM.fundamentalDomain`

Definition 1.3 (Hamiltonian vector fields, symplectic form)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $X_H(z) = (\partial_I H(z), -\partial_\theta H(z))$; for holomorphic h on an open subset of $\mathbb{C}^n \times \mathbb{C}^n$, X_h is defined by the same formula with complex derivatives. The standard symplectic form is $\omega_0((a,b),(c,d)) = \sum_i (b_i c_i - a_i d_i)$, i.e. $\omega_0 = \sum_i dI_i \wedge d\theta_i$, and $\Omega_0(a,b) = (-b,a)$ is its matrix $\left(\begin{smallmatrix} 0 & -I \\ I & 0 \end{smallmatrix}\right)$.
Frozen Lean declaration: `KAM.hamiltonianVectorField, KAM.hamiltonianVectorField$\mathbb{C}$, KAM.symplecticForm, KAM.omegaZero`
Source: HL19 Rem. 2.1 and §2.2, p. 6.

Remark 1.4 (Lean totality). The frozen definitions use `fderiv`, which is 0 where a function is not differentiable, the matrix inverse, which is 0 for a singular matrix, and the Bochner integral, which is 0 for a non-integrable function. Every frozen theorem guards these by hypotheses (differentiability on the relevant open set, invertibility, integrability through continuity); the proofs below must establish the guarded facts before using the defining formulas.

Definition 1.5 (The integrable part, the perturbation, the frequency map)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$, the real Hamiltonian is $H(\theta, I) = \mathrm{Re}\big(h(I) + f(\theta, I)\big)$ at real points, and the frequency map is $\nabla h(I)$, the real gradient of $I \mapsto \mathrm{Re}\, h(I)$ at real I.
Frozen Lean declaration: `KAM.realHamiltonian, KAM.frequencyMap`
Source: Pos01, Classical KAM Theorem, p. 6.

Definition 1.6 (Invariant Lagrangian torus)**.** A map $K : \mathbb{R}^n \to \mathbb{R}^n \times \mathbb{R}^n$ is an *invariant Lagrangian torus of H with frequency ω* if (i) K is real-analytic; (ii) $K(\theta + k) = K(\theta) + (k, 0)$ for $k \in \mathbb{Z}^n$ (primary torus); (iii) $K(\theta') = K(\theta) + (k, 0)$ implies $\theta' = \theta + k$; (iv) $DK(\theta)$ is injective; (v) $\omega_0(DK(\theta)u, DK(\theta)v) = 0$ for all u, v; (vi) for all θ and t, $\frac{d}{dt} K(\theta + t\omega) = X_H(K(\theta + t\omega))$.
Frozen Lean declaration: `KAM.IsInvariantLagrangianTorus`
Source: HL19 (2.1), (2.3), Rem. 2.2–2.3, pp. 6–7; ruling Q3.

Definition 1.7 (The KAM cover)**.** Given $h, f, B, r_0, \tau, \delta$, the set $\mathcal{K}(\delta)$ consists of the points $z \in K(\mathbb{R}^n)$ for some K, ω, γ, I_0 with: $\gamma > 0$; $\omega \in \mathcal{D}_{\gamma,\tau}$; K an invariant Lagrangian torus of H with frequency ω; every action $K(\theta)_2$ within r_0 of B; $I_0 \in B$ and $\nabla h(I_0) = \omega$; and $|K(\theta) - (\theta, I_0)| \le \delta$ for all θ.
Frozen Lean declaration: `KAM.kamCover`

Definition 1.8 (Torus data on strips)**.** Let $K : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$, h holomorphic, $\omega \in \mathbb{R}^n$. K is *primary on T_ρ* if $K(\theta + k) = K(\theta) + (k, 0)$ for $\theta \in T_\rho$, $k \in \mathbb{Z}^n$, and *real on reals* if $K(\mathbb{R}^n) \subset \mathbb{R}^n \times \mathbb{R}^n$. The invariance error is $E(\theta) = X_h(K(\theta)) - DK(\theta)\omega$. With $L = DK$ (a $2n \times n$ matrix) and the bilinear transpose, the Gram matrix is $L^\top L$, $B = (L^\top L)^{-1}$, the normal frame is $N = \Omega_0 L B$, $X_N = DX_h(K)N - DN[\omega]$, the torsion is $T = N^\top \Omega_0 X_N$, and $\langle T \rangle = \int_{[0,1]^n} T(\theta)\, d\theta$.

Frozen Lean declaration: `KAM.IsPrimaryOnStrip`, `KAM.IsRealOnReals`, `KAM.invarianceError`, `KAM.dotℂ`, `KAM.gramMatrix`, `KAM.normalFrame`, `KAM.torsion`, `KAM.torsionAverage`
Source: HL19 (2.2), (3.2)–(3.9), pp. 7, 16–17; Rem. 2.3, 3.3 (canonical Case III, $A = 0$).

1.3. **The frozen theorems.**

Theorem 1.9 (HEAD: positive-measure KAM with rate)**.** *Let $n \geq 2$, $\tau > n-1$, $s_0, r_0 > 0$, and $B \subset \mathbb{R}^n$ open and bounded with $\operatorname{meas}\{I \in B : \operatorname{dist}(I, B^c) < \delta\} \leq C_B\delta$ for all $\delta > 0$. Let h be holomorphic on B_{r_0}, real on real points, with $|I - J| \leq L|\nabla h(I) - \nabla h(J)|$ on $\bar{B}$. Then there are $C, \varepsilon_0 > 0$ such that for every f holomorphic on $D(s_0, r_0)$, 1-periodic in θ, real on real points, and every $\varepsilon \leq \varepsilon_0$ with $|f| \leq \varepsilon$ on $D(s_0, r_0)$:*

$$\operatorname{meas}\big(F_B \setminus \mathcal{K}(C\sqrt{\varepsilon})\big) \leq C\varepsilon^{1/4}.$$

Frozen Lean declaration: `KAM.kam_positive_measure`
Source: Pos01, Classical KAM Theorem, p. 6, at the rate of decision D8 (ledger D-MEAS-1).

Theorem 1.10 (HEAD-Q: qualitative form)**.** *Same data as Theorem 1.9 without the boundary condition. For every $\kappa > 0$ there is $\varepsilon_0 > 0$ such that, for f, ε as there with $\varepsilon \leq \varepsilon_0$, $\operatorname{meas}(F_B \setminus \mathcal{K}(\kappa)) \leq \kappa \operatorname{meas}(F_B)$.*
Frozen Lean declaration: `KAM.kam_positive_measure_qualitative`

Theorem 1.11 (G1: a-posteriori KAM at fixed frequency)**.** *Let $n \geq 2$, $\tau \geq n-1$, $0 < \rho_\infty < \rho$, $M, \sigma_K, \sigma_B, \sigma_T, \mu, \bar{\gamma} > 0$. There are $C_1, C_2 > 0$ such that the following holds for all $\mathfrak{B}, h, K, \omega, \gamma, e$. Hypotheses: $\mathfrak{B} \subset \mathbb{C}^n \times \mathbb{C}^n$ open and invariant under $\theta \mapsto \theta + k$; h holomorphic on $\mathfrak{B}$, 1-periodic in θ, real on real points, with $|Dh|, |X_h|, |DX_h|, |D^2X_h| \leq M$ on $\mathfrak{B}$; K holomorphic on T_ρ, primary, real on reals, with $B_\mu(K(\theta)) \subset \mathfrak{B}$ and $|DK(\theta)| \leq \sigma_K - \mu$ on T_ρ; $L^\top L$ invertible with $|B| \leq \sigma_B - \mu$ on T_ρ; $\langle T \rangle$ invertible with $|\langle T \rangle^{-1}| \leq \sigma_T - \mu$; $0 < \gamma \leq \bar{\gamma}$ and $\omega \in \mathcal{D}_{\gamma,\tau}$; $|E| \leq e$ on T_ρ and $C_1 e/(\gamma^4 \rho^{4\tau}) < 1$. Conclusion: there is K_∞, holomorphic on T_{ρ_∞}, primary, real on reals, with $E_\infty = 0$ on T_{ρ_∞}, $B_{\mu/2}(K_\infty(\theta)) \subset \mathfrak{B}$, $|DK_\infty| \leq \sigma_K - \mu/2$, and $|K_\infty - K| \leq C_2 e/(\gamma^2 \rho^{2\tau})$ on T_{ρ_∞}.*
Frozen Lean declaration: `KAM.aposteriori_kam`
Source: HL19 Theorem 3.1, pp. 16–18 (ledger D-G1-1).

Theorem 1.12 (G2: local uniqueness for flows)**.** *Let $n \geq 2$, $\tau \geq n-1$, $\rho > 0$, $M, \sigma_K, \sigma_B, \sigma_T, \mu > 0$. There is $c > 0$ such that: if $\mathfrak{B}, h$ are as in G1, $\omega \in \mathcal{D}_{\gamma,\tau}$ with $\gamma > 0$, K_1 satisfies the hypotheses of G1 on T_ρ (with these constants) and $E_1 = 0$, K_2 is holomorphic on T_ρ, primary, real on reals, $K_2(T_\rho) \subset \mathfrak{B}$, $E_2 = 0$, and $|K_1 - K_2| \leq c\gamma^2\rho^{2\tau}$ on T_ρ, then $K_2(\theta) = K_1(\theta + \varphi)$ on $T_{\rho/2}$ for some $\varphi \in \mathbb{R}^n$.*
Frozen Lean declaration: `KAM.local_uniqueness_flow`
Source: Our own result (ledger D-UNIQ-1); template GJLV05 Theorem 2, p. 862.

Theorem 1.13 (G3: measure of the non-Diophantine set)**.** *Let $\tau > n-1$ and $\Omega \subset \mathbb{R}^n$ bounded. There is C such that $\operatorname{meas}\{\omega \in \Omega : \omega \notin \mathcal{D}_{\alpha,\tau}\} \leq C\alpha$ for every $\alpha > 0$.*
Frozen Lean declaration: `KAM.volume_not_diophantine_le`
Source: Pos01 §1d, p. 5.

2. Analysis on strips and complex domains

Status: interface fixed by the orchestrator (27 Oct 2026); proofs to be written. The other parts cite only the labels below. A writer of another part who needs a further infrastructure lemma states it in that part, as a lemma of its own, and lists it for transfer here.

Throughout, functions on strips are holomorphic on the open strip and bounded there, and norms are sup norms (Part 1). Constants called $c(\cdot)$ depend only on the displayed quantities.

2.1. **Cauchy estimates.**

Lemma 2.1 (Cauchy estimate on strips)**.** *Let u be holomorphic and bounded on T_ρ and $0 < \delta < \rho$. Then for every ℓ, $\|\partial_{\theta_\ell} u\|_{\rho-\delta} \le \delta^{-1}\|u\|_\rho$, hence $\|Du\|_{\rho-\delta} \le n\delta^{-1}\|u\|_\rho$; the same holds for vector- and matrix-valued u with the norms of Part 1.*
Source: HL19 §3.1, p. 15.

Lemma 2.2 (Cauchy estimate on open sets)**.** *Let f be holomorphic on an open $U \subset \mathbb{C}^m$ with $|f| \le A$ on U, and let the closed polydisc of radius δ around z lie in U. Then $|\partial_\ell f(z)| \le A/\delta$, and $|D^j f(z)| \le c(m,j)A\delta^{-j}$ for $j = 1, 2, 3$.*

2.2. **Fourier coefficients of periodic holomorphic functions.**

Lemma 2.3 (Coefficient decay)**.** *Let u be holomorphic, bounded and 1-periodic on T_ρ. Then $|\hat{u}_k| \le e^{-2\pi|k|_1\rho'}\|u\|_\rho$ for every $k \in \mathbb{Z}^n$ and $0 \le \rho' < \rho$.*

Lemma 2.4 (Summation)**.** *If $\sum_k |a_k| e^{2\pi|k|_1\rho} < \infty$, then $u(\theta) = \sum_k a_k e^{2\pi i k\cdot\theta}$ is holomorphic and 1-periodic on T_ρ, $\hat{u}_k = a_k$, and $\|u\|_\rho \le \sum_k |a_k| e^{2\pi|k|_1\rho}$. Conversely, for u as in Lemma 2.3 and $0 < \delta < \rho$, $\sum_k |\hat{u}_k| e^{2\pi|k|_1(\rho-\delta)} \le \coth(\pi\delta)^n \|u\|_\rho$, and u equals its Fourier series on T_ρ.*

2.3. **The cohomological equation.**

Lemma 2.5 (Cohomological equation on strips)**.** *Let $\omega \in \mathcal{D}_{\gamma,\tau}$ with $\gamma > 0$, $\tau \ge 0$, and let g be holomorphic, bounded and 1-periodic on T_ρ with $\langle g \rangle = 0$. For $0 < \delta < \rho$ there is a unique u, holomorphic and 1-periodic on $T_{\rho-\delta}$, with $\langle u \rangle = 0$ and $Du(\theta)\omega = g(\theta)$, and*

$$\|u\|_{\rho-\delta} \le c(n,\tau)\,\gamma^{-1}\delta^{-(\tau+n)}\|g\|_\rho.$$

If g is real on real points, so is u. The same holds componentwise for vector- and matrix-valued g.
Source: HL19 §4.1 (with Rüssmann's sharper exponent τ; the cruder exponent $\tau + n$ suffices, decision D12). Coefficient form: `docs/cohomological-equation.md`, Lean `KAM.Diophantine`.

Lemma 2.6 (Averages)**.** *For v holomorphic, bounded and 1-periodic on T_ρ: $\langle \partial_{\theta_\ell} v \rangle = 0$, $\langle Dv\,\omega \rangle = 0$, and $|\langle v \rangle| \le \|v\|_\rho$.*

2.4. **Matrices.**

Lemma 2.7 (Neumann series)**.** *If A is invertible and $|A^{-1}||\Delta| \le \frac{1}{2}$, then $A + \Delta$ is invertible with $|(A+\Delta)^{-1}| \le 2|A^{-1}|$ and $|(A+\Delta)^{-1} - A^{-1}| \le 2|A^{-1}|^2|\Delta|$. The same holds pointwise for matrix-valued functions, with sup norms.*

2.5. **Diophantine frequencies.**

Lemma 2.8 (Measure of the resonant set)**.** *This is Theorem 1.13; its proof is written in Part 5.*

3. The a-posteriori theorem at fixed frequency (G1)

Status: being written.

4. Local uniqueness of invariant tori of flows (G2)

Status: being written.

5. The measure of the invariant tori (HEAD, HEAD-Q, G3)

Status: being written.